

Business Model Canvas: AIIA Clinical Trials Dashboard

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1. Purpose

This document expands the **Business Model Canvas** for the AIIA Clinical Trials Dashboard into a standalone, detailed artefact. The nine building blocks are set out in full below, each with its content, the reasoning behind it, and the specific risk it carries.

The canvas describes how the idea **earns and spends**. The revenue arithmetic behind it is developed separately in the **Business Perspective** document.

2. Canvas at a glance

#	Block	One-line summary
1	Key Partners	AYUSH institutes, IECs and NPvCCs, CTRI and ABDM
2	Key Activities	Trial tracking, safety capture, audit trail, rule-engine upkeep
3	Key Resources	ASU rule engine, FHIR schema, nine role portals, deployed runtime
4	Value Proposition	One live record per study, compliance built in, zero licence fee
5	Customer Relationships	Institution licence, PI-led adoption, guided onboarding
6	Channels	Direct to institute, AIIA network, conferences, grant-funded pilots
7	Revenue Streams	Subscription, per-study pricing, setup and training, pilot fees
8	Customer Segments	AYUSH colleges and hospitals, research institutes, NPvCC/IPvCC, pharma and CROs
9	Cost Structure	Serverless hosting, ~3 dev-days/month, training and support

3. Block 1 — Key Partners

AYUSH institutes, colleges and hospitals — the primary buyers and the institutions whose trial operations the system replaces. They are also the gatekeepers of adoption: without institutional sponsorship, clinicians at that site will not use the dashboard.

Ethics Committees (IEC) and Pharmacovigilance Centres (NPvCC/IPvCC) — approval and safety-reporting bodies whose timelines the system enforces. Partnership here is not merely commercial; their sign-off is required for compliance mapping to be accepted.

Clinical Trials Registry – India (CTRI) and ABDM — the registries against which study status is validated. No usable public API exists, so the working relationship today is CSV and FHIR bulk import, with live sync contingent on a formal MoU.

Risk: every partner is a public-sector body with its own procurement and approval cycles. None of these relationships can be assumed; each requires institutional buy-in that the product team does not control.

4. Block 2 — Key Activities

Trial tracking — enrolment, visit scheduling and query management against a single record, so status is a query rather than a compilation task.

Safety capture — AE/SAE entry and routing on regulatory timelines, with clock-based automatic escalation.

Immutable audit trail — maintaining the time-stamped, attributable record that makes an audit a query rather than a reconstruction exercise.

Regulatory rule-engine upkeep — encoding GCP-ASU, NDCT Rules 2019, CTRI status and pharmacovigilance obligations as executable rules, and keeping them current as regulation changes. This is the differentiating activity and it is continuous, not one-off.

Onboarding and training — migration, role mapping and shadowing sessions that determine whether adoption succeeds.

Risk: rule-engine upkeep is ongoing regulatory work disguised as engineering. It will not scale by adding features; it scales by adding domain experts.

5. Block 3 — Key Resources

The ASU regulatory rule engine — the core differentiator. Global CTMS products encode ICH-GCP pharma workflows; this encodes Ayurveda regulation. It is the reason the audit trail is produced automatically rather than assembled manually.

The FHIR R4 data schema — makes Ayurveda trial data structurally comparable with global health data, which is what allows interoperability without a bespoke mapping project.

Nine role portals — investigator, coordinator, ethics committee, pharmacovigilance, institutional leadership, participant, regulator/auditor and administrative roles, each with least-privilege data scope and its own KPI set.

The deployed runtime — a serverless edge stack (Cloudflare Workers, D1, R2) with a typed end-to-end application (TypeScript, React 19, Next.js, Drizzle) and PBKDF2 authentication via WebCrypto.

Risk: the rule engine and schema are the assets a competitor would need to replicate. They are valuable but they are not proprietary in any legal sense, and once demonstrated the advantage is execution speed rather than exclusivity.

6. Block 4 — Value Proposition

For AYUSH institutions: a clinical-trial management system that understands Ayurveda regulation, at a fraction of the cost of a global CTMS — replacing spreadsheets with an auditable system of record.

For ethics committees: approval expiry and deviations surfaced before they become audit findings, so corrective action happens in days rather than months.

For pharmacovigilance: AE and SAE records routed on regulatory timelines instead of chased over email.

For participants: their own study, visits and ID card in a portal a non-specialist can actually read.

For regulators and auditors: a complete, immutable ALCOA+ trail instead of reconstructed files.

The core claim: compliance is not a checklist to be verified after the fact. GCP-ASU, CTRI and NDCT rules are built into the software, so compliance is a property of the system rather than an activity performed by staff.

Risk: the value proposition depends on a regulatory mapping that has not yet been independently validated or audited. Until it is, this is a design claim, not a certified one.

7. Block 5 — Customer Relationships

Institution licence with guided onboarding — a subscription relationship rather than a transaction, with setup, data migration, role mapping and shadowing sessions included.

PI-led adoption — adoption is driven by principal investigators who champion the tool within their institution, not by a central mandate. This is the highest-signal channel in an institution where senior clinicians resist new screens.

Adoption tracked as a KPI — active users against named roles is monitored as a business metric, not merely an operational one, because adoption is the dominant risk to the entire model.

Risk: PI-led adoption is also PI-dependent. When a champion moves institutions, the relationship can lapse. Institutional contracts with onboarding commitments mitigate this but cannot eliminate it.

8. Block 6 — Channels

Direct to institute — direct outreach to AYUSH colleges, hospitals and research institutes, supported by a live prototype that can be demonstrated end-to-end without an installation.

Allia network — institutional relationships around the All India Institute of Ayurveda and the wider Ministry of Ayush network, which is the single most concentrated route to credibility for an ASU-specific product.

Conferences and consortia — AYUSH academic and professional conferences, and institutional research consortia.

Grant-funded pilots — the primary entry route: a pilot costs the institution little, produces an adoption result the team can evidence, and converts to a paid licence on success.

Risk: grant-funded pilots consume support time disproportionately. If pilots do not convert, the channel becomes a loss-making services business. Pilot-to-paid conversion is therefore the single most important metric in the whole model.

9. Block 7 — Revenue Streams

Stream	Price	Notes
Institution licence	₹1,50,000–3,00,000 / year	Recurring subscription; primary revenue
Per active study	₹15,000–30,000 / study / year	Usage-linked upside
Setup and training	₹50,000 one-time, plus ₹1,000/person/day thereafter	Covers migration and onboarding effort
Grant-funded pilot fees	Discounted or deferred	Acquisition channel rather than profit centre

Note on the deck. The pitch presentation currently presents ₹2,500–6,000 per month as the software cost, which is the platform hosting cost, not a price. The pricing above should replace it, with hosting shown separately as a cost of goods.

Risk: institutional budgets in the AYUSH sector are often grant-cycle dependent. A ₹2,00,000 annual subscription competing with free spreadsheets is a real decision, and it is made by people who did not build the tool and may not use it daily.

10. Block 8 — Customer Segments

AYUSH colleges and hospitals — the core segment. Institutional mandate to conduct Ayurveda research, constrained budgets, and no existing compliant tool.

AYUSH research institutes — higher regulatory maturity and stronger compliance motivation, and therefore the earliest credible adopters.

NPvCC and IPvCC — pharmacovigilance centres that need reconciled safety data across many studies; natural multi-tenant customers.

Pharma companies and CROs running ASU trials — the highest-value segment with the strongest purchasing process and the longest sales cycle.

Segment prioritisation: research institutes and NPvCCs should be targeted first, because their compliance need is strongest and their decision cycle shortest. Pharma and CROs are the most valuable segment but should be treated as a later phase, not the beachhead.

Risk: treating pharma and CROs as the primary segment would misrepresent the product's current readiness. The system has no validated integration to any sponsor or CRO platform.

11. Block 9 — Cost Structure

Serverless hosting — ₹2,500–6,000 per month for the full platform (Cloudflare Workers, D1, R2). Effectively zero marginal cost per additional institution.

Engineering maintenance — approximately 3 developer-days per month to operate the deployed system, per the current feasibility assessment.

Training and support — onboarding, shadowing sessions and helpdesk; the largest controllable cost and the one that scales with institutions, not with users.

Compliance and validation work — control-to-clause mapping, IEC sign-off, audit-trail evidence. This is legal and documentation effort rather than code, and it is what converts a prototype into something procurable.

Cost characterisation: the model has a very low fixed cost base and near-zero marginal cost, which places break-even at approximately one institution. That excludes founder compensation, validation cost and statutory compliance, and should be described as product sustainability rather than profitability.

Risk: training and support are labour-bound and do not benefit from the serverless economics that make hosting free. Growth in institutions therefore grows cost in a way the gross-margin figures can obscure.

12. Canvas-level risks

1. **Adoption is the binding constraint.** Every block assumes clinicians adopt a new screen. This is the one risk that cannot be engineered away.
2. **Compliance mapping is unvalidated.** The central value proposition depends on a regulatory mapping that no external body has yet reviewed.
3. **Partner dependency.** All key partners are public institutions whose timelines the team does not control.
4. **Pilot conversion unproven.** The channel most relied upon has no conversion data.
5. **Cost model omits people.** The low cost base excludes founder and validation cost, so break-even is optimistic.
6. **Single-vendor runtime.** Continuity depends on one serverless cloud provider.

13. Conclusion

The canvas describes a coherent, deliberately simple business: partner with the institutions that are mandated to run Ayurveda trials, encode the regulation those institutions are accountable for, sell

access to a single auditable record at a price their budgets can absorb, and let near-zero marginal cost carry the economics. Its strength is that the regulatory rule engine creates a genuine barrier that generic CTMS vendors have no reason to build. Its weakness is that every block depends on institutional adoption and partner cooperation, neither of which the team controls — which is why adoption, not market size, is the number to watch.

14. References

- Clinical Trials Registry – India — ctri.nic.in
- Ayush Suraksha — suraksha.ayush.gov.in
- GCP / ICMR-National Ethical Guidelines (CDSR) — cdsr.nic.in
- NDCT Rules 2019 — indiacode.nic.in
- DPDP Act 2023 (Meity) — meity.gov.in
- HL7 FHIR R4 — hl7.org/fhir
- CDISC — cdisc.org
- All India Institute of Ayurveda, Ministry of Ayush — aiia.gov.in
- Project artefacts: prototype — aiia-research-observatory-demo.vercel.app · source — github.com/Amirthaa1810/aiia-research-observatory

Status: Business Model Canvas for review, to be read alongside the Business Perspective document.